Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/25/2017
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Utilizing the Advanced Biological Research System (ABRS) on the International Space Station (ISS) to Uncover Microgravity's Impact on Root Development and Cell Wall Architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/25/2012	**End Date:**	07/24/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 7/24/2017 per PI and NSSC information (Ed., 7/13/16) NOTE: Extended to 7/24/2016 per PI and NSSC information (Ed., 5/27/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nakashima, Jin (Noble Foundation) Tang, Yuhong (Noble Foundation)		
Grant/Contract No.:	NNX12AM94G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this proposal is to uncover the underlying molecular mechanisms by which microgravity impacts cell wall architecture in the model plant Arabidopsis thaliana using the Advanced Biological Research System (ABRS) housed on the International Space Station (ISS). This proposal builds on a robust dataset from our rapid turnaround spaceflight experiments, which utilized the Biological Research in Canisters (BRIC16) hardware on the STS-131 mission. Through experiments with the BRIC hardware, we discovered that spaceflight had a profound influence on the expression of genes involved in regulating cell wall architecture. This led us to hypothesize that plants respond to microgravity in large part through the transcriptional reprogramming of genes that control cell wall remodeling. Because of the prominent root developmental phenotypes (i.e., primary root skewing and inhibition of root hair growth) that we observed in our BRIC experiments, it appears that microgravity-dependent changes in cell wall-related transcripts are predominantly manifested in roots. Furthermore, because primary root and root cell wall ultrastructural defects were most pronounced in a mutant to a vegetative actin isoform (act2), we hypothesize that microgravity predominantly affects genes that lie within signaling networks linking the actin cytoskeleton with the plant extracellular matrix. Given our successful experiments with BRIC, we are in an excellent position to effectively utilize the unique capabilities of the ABRS hardware on the ISS to test the aforementioned hypotheses. We will use a new algorithm that we developed from our BRIC16 experiments to more rigorously quantify root growth behavior of wild-type and a higher order vegetative actin mutant (act2 act8 double mutant) (aim 1). This will clarify the extent of actin involvement in specifying root orientation and cell wall remodeling in space. Through the use of monoclonal antibodies and immunocytochemistry, we will also determine how the distribution and abundance of specific cell wall components change in microgravity, and correlate such changes with root developmental responses and modifications in gene expression (aim 2). Finally, we will employ more sensitive deep sequencing technologies (e.g., RNA-Seq) to discover novel microgravity-induced transcriptional changes in plants (aim 3). Knowledge from our spaceflight studies will then be translated into corresponding gene functional studies (e.g., mutant analysis and overexpression), which we expect will lead to the discovery of new players that modulate root development on Earth and in microgravity. The proposed work is therefore relevant to this NASA-NRA (NASA Research Announcement) because it will address fundamental questions on how gravity modulates the plant's internal cytoarchitecture and the molecular processes that govern the development of its cell walls. This will be significant for NASA's mission as extrapolated in the priorities of the Fundamental Space Biology Plan for 2010-2020 because it will provide us with basic knowledge to guide genetic engineering efforts to generate plants that are better adapted to the weightless environment of spaceflight. This in turn will be beneficial for the design of large plant habitats on the ISS and future space colonies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide a deeper understanding of how microgravity influences plant development. The focus of this research will be on the development of roots and how they modify the expression of their genes to adapt to the microgravity environment of spaceflight. This research will address one of high priority recommendations of the National Research Council’s 2011 decadal survey report, which is to contribute significantly to building a robust spaceflight program to analyze plant physiological responses to stimuli encountered in microgravity. Research areas depicted in this proposal, namely root development and cell wall biology, are significant for NASA’s goal of developing large plant habitats on the International Space Station (ISS) and future space colonies. We expect that information gained from basic studies of root growth in microgravity will guide the development of plant varieties better equipped to optimize nutrient and water acquisition, and waste recycling in enclosed regenerative life support systems. Furthermore, because upright plants evolved under the 1-g conditions of Earth in large part through the mechanical strength of their cell walls, this research will provide a deeper mechanistic understanding of the genetic networks that specify how cell walls are constructed in microgravity. In space, such knowledge will enable technologies for designing plants that can sustain optimal growth and stature to overcome the adverse impact that microgravity has on cell wall development. With regard to Earth benefits, understanding of molecular pathways that govern root growth and cell wall development, will benefit agricultural practices and bioenergy research on Earth.
Task Progress:	We achieved all of the major milestones we set for this proposal. Malfunction of the ABRS hardware on the ISS, de-manifestation from Space X-3, and delays in the launch of Space X-5 were set-backs we encountered during the first two years of the grant. However, we were able to overcome these early problems and completed most of the specific aims that we proposed. A major accomplishment of this proposal was the implementation of a simple method to delay germination of Arabidopsis seeds until experimental activation in space. We foresee that many in the plant space biology community will benefit from these methods. Another direct outcome of this grant is the discovery of novel plant proteins involved in modulating the cytoskeleton. These results have been published in high impact peer-reviewed publications. We also successfully retrieved material from ground- and space-grown Arabidopsis seedlings. Processing and data analysis from these spaceflight experiments have been completed. We have deposited raw RNA-Seq data from our spaceflight work to the NASA GeneLab site but these have not yet been publicly released. During this funding period, we also completed glycomic studies of Arabidopsis seedlings grown in space and compared them with the corresponding ground controls. Both the transcriptomics (i.e., RNA-Seq) and glycomics results will be submitted this year for peer-review. Taken together, our RNA-Seq and glycomics results continue to point to the spaceflight environment as a major trigger for oxidative stress responses in plants. Our results suggest that the ideal plant ideotype for future space habitats could be similar to those that have been developed for marginal agricultural lands on Earth including crops tolerant to low oxygen (e.g., hypoxia) and other stressors (drought, insect resistance) as well as those with more robust cell wall mechanical properties. Such findings could be the bases of future spaceflight experiments wherein performance of plants with enhanced tolerance to abiotic stress can be evaluated on hardware such as the advanced plant habitat (APH) or the Spectrum.
Bibliography Type:	Description: (Last Updated: 01/13/2026)

Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Blancaflor EB. "A forward genetic screen on chemicals that disrupt the actin cytoskeleton uncovers a novel regulator of auxin efflux carrier trafficking in Arabidopsis." Presented at the 2016 International Conference on Plant Growth Substances, Toronto, Canada, June 21-25, 2016. 2016 International Conference on Plant Growth Substances, Toronto, Canada, June 21-25, 2016. , Jun-2016
Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Renna L, Brandizzi F, Blancaflor EB. "Identification of a novel tetratricopeptide repeat domain-containing protein that mediates actin-endomembrane crosstalk in plants." Presented at Plant Biology 2016, Austin, Texas, July 9-13, 2016. Plant Biology 2016, Austin, Texas, July 9-13, 2016. , Jul-2016
Abstracts for Journals and Proceedings	de Bang L, Blancaflor EB. "Brassinosteroids regulate organ straightening during gravitropism through modification of actin filaments." Presented at the 2016 Meeting of the Southern Section of the American Society of Plant Biologists, Denton, Texas April 2 – 4, 2016. 2016 Meeting of the Southern Section of the American Society of Plant Biologists, Denton, Texas April 2 – 4, 2016. , Apr-2016
Abstracts for Journals and Proceedings	Yoo C-M, Sparks JA, Blancaflor EB. "Class 1 ARF-GAPs: regulatory hubs that link phosphoinositide signaling and the cytoskeleton in specifying root hair polarity." Presented at the 9th International Symposium on Root Research, Canberra, Australia, October 6-9, 2015. 9th International Symposium on Root Research, Canberra, Australia, October 6-9, 2015. , Oct-2015
Abstracts for Journals and Proceedings	Nakashima J, Avci U, Pattathil S, Sparks JA, Kwon T, Hahn MG, Blancaflor EB. "Glycomic profiling combined with immunocytochemistry reveals microgravity-induced changes in cell walls of Arabidopsis seedlings." Presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Kwon T, Khan BR, Sparks JA Blancaflor EB. "Hypersensitive to Latrunculin B (HLB2) encodes a BEACH-domain protein that functions in epidermal cell expansion." Presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Kwon T, Sparks JA, Nakashima J, Allen SN, Krom N, Tang Y, Blancaflor EB. "Transcription profiles of Arabidopsis thaliana grown in space reveal regulatory networks for the adaptive response to microgravity." Presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Renna L, Brandizzi F, Blancaflor EB. "Hypersenstive to Latrunculin B 1 mediates crosstalk between the actin cytoskeleton and endomembranes during root development in Arabidopsis." Presented at the 8th International Symposium on Root Development, Umea, Sweden May 29- June 1, 2017. 8th International Symposium on Root Development, Umea, Sweden May 29- June 1, 2017. , May-2017
Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Khan BR, Blancaflor EB. "Identification of novel proteins involved in cytoskeleton-endomembrane crosstalk through forward genetic screens on actin disrupting compounds." Presented at the 28th International Conference on Arabidopsis Research, St. Louis, Missouri, June 19-23, 2017. 28th International Conference on Arabidopsis Research, St. Louis, Missouri, June 19-23, 2017. , Jun-2017
Articles in Peer-reviewed Journals	Yoo CM, Naramoto S, Sparks JA, Khan BR, Nakashima J, Fukuda H, Blancaflor EB. "Deletion analysis of AGD1 reveals domains crucial for plasma membrane recruitment and function in root hair polarity." Journal of Cell Science. 2018 Jan 29;131(2):jcs2038282017. [Epub 2017 Jun 23] http://doi.org/10.1242/jcs.203828 ; PubMed PMID: 28646092 , Jan-2018
Articles in Peer-reviewed Journals	Kwon T, Sparks JA, Liao F, Blancaflor EB. "ERULUS is a plasma membrane-localized receptor-like kinase that specifies root hair growth by maintaining tip-focused cytoplasmic calcium oscillations." Plant Cell. 2018 Jun;30(6):1173-7. Epub 2018 May 25. https://doi.org/10.1105/tpc.18.00316 ; PubMed PMID: 29802213; PubMed Central PMCID: PMC6048781 , Jun-2018
Articles in Peer-reviewed Journals	Nakashima J, Pattathil S, Avci U, Chin S, Alan Sparks J, Hahn MG, Gilroy S, Blancaflor EB. "Glycome profiling and immunohistochemistry uncover changes in cell walls of Arabidopsis thaliana roots during spaceflight." npj Microgravity. 2023 Aug 22;9:68. https://doi.org/10.1038/s41526-023-00312-0 , Aug-2023
Books/Book Chapters	Paez-Garcia A, Sparks JA, De Bang L, Blancaflor EB. "The plant actin cytoskeleton: new functions from an old scaffold." in "Concepts in Cell Biology - History and Evolution. Plant Cell Monographs, vol 23. " Ed. V. Sahi, F. Baluska. Cham: Springer-Verlag, 2018. p. 103-137. https://doi.org/10.1007/978-3-319-69944-8_6 , Mar-2018
Journal/Magazine covers	Yoo CM, Naramoto S, Sparks JA, Khan BR, Nakashima J, Fukuda H, Blancaflor EB. "Cover image in Journal of Cell Science for article, 'Deletion analysis of AGD1 reveals domains crucial for its plasma membrane recruitment and function in root hair polarity.' " J Cell Sci. 2018 Jan 29;131(2):jcs203828. PubMed PMID: 28646092 ; http://jcs.biologists.org/content/131/2.cover-expansion , Jan-2018